TCC 2026

Relaxed Modular PCS from Arbitrary PCS

and Applications to SNARKs for Integers

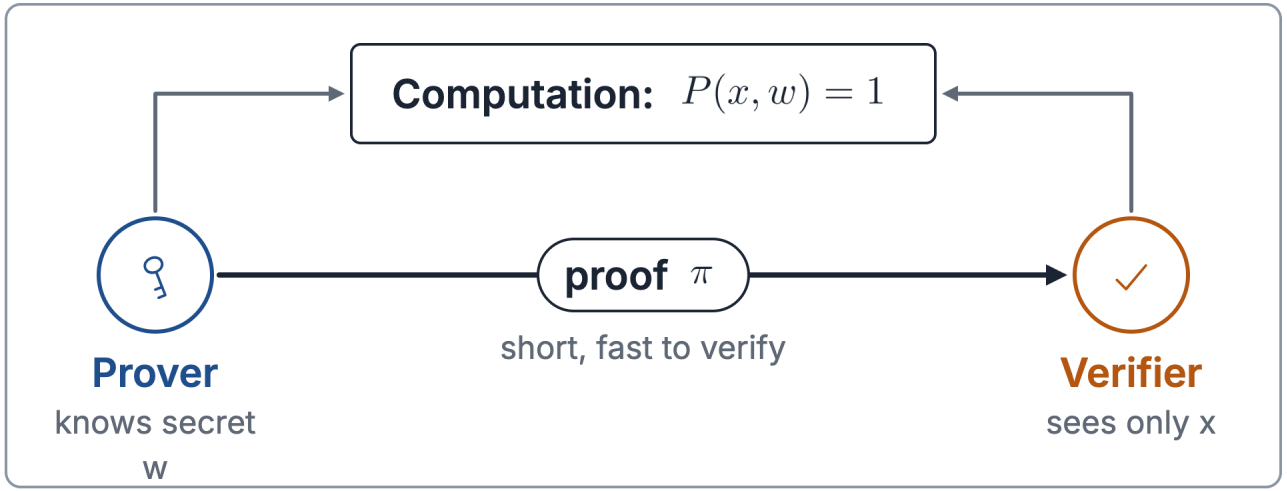
Alireza Shirzad* · Sriram Sridhar* · Dimitrios Papadopoulos · Charalampos Papamanthou

University of Pennsylvania · UC Berkeley · HKUST · Lagrange Labs and Yale University

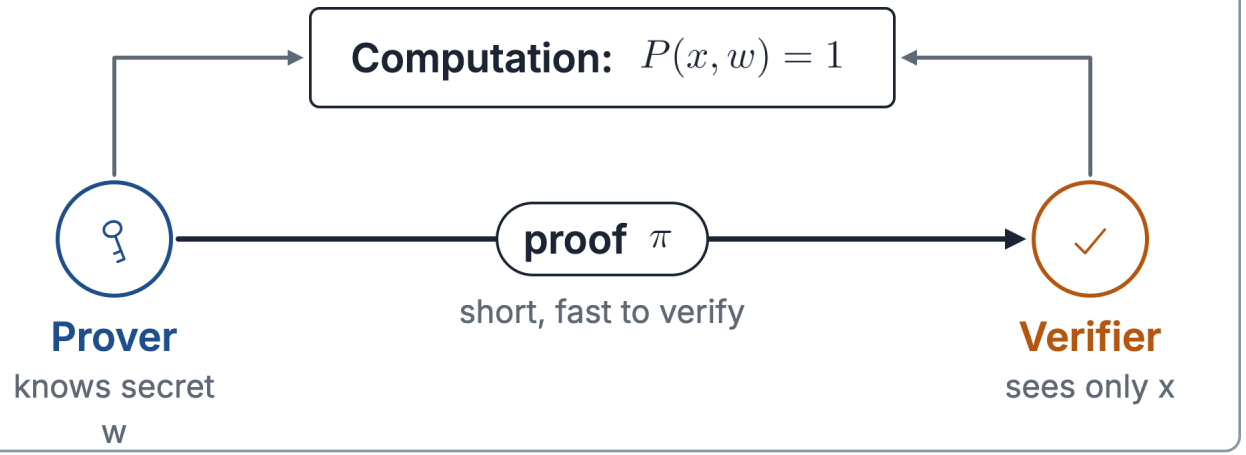
* Work partially done while at Lagrange Labs

These slides were created by AI under human supervision.

SNARKs and the computations they prove



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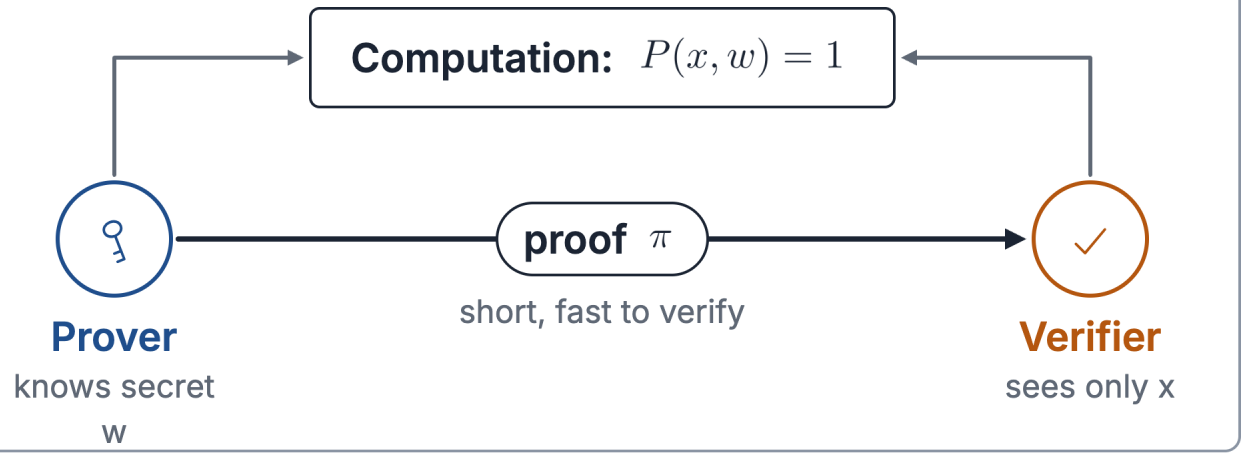


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The computation P is usually expressed in an NP-complete language over a finite field.

Examples: R1CS, circuit satisfiability, Plonkish.

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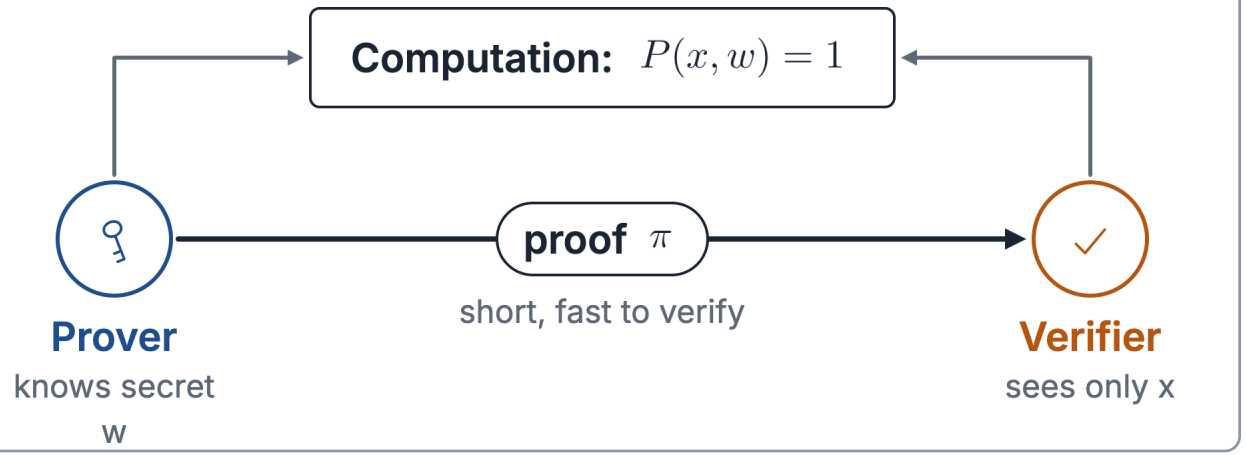
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Machine learning floating and fixed point

CPU operations the ring $\mathbb{Z}/2^k\mathbb{Z}$ k-bit words

What we want to prove

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The gap

Finite-field languages
R1CS, circuit satisfiability, Plonkish

What we want to prove | **How P is expressed**

The price of emulation inside a field

A huge blowup

$2^5\times$

for basic 32-bit CPU arithmetic

$> 2^{12}\times$

for cryptographic computations such as RSA

RSA and bug figures as reported in Limber, CXNB26

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Soundness pitfalls

Every range check and carry relation is written by hand. One omission breaks soundness.

9 of 27 critical bugs in real-world ZK systems came from non-native arithmetic

$450k bug bounty paid for one non-native arithmetic soundness bug

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RSA: one constraint per multiplication, $a \cdot b \equiv c \pmod{N}$

For two RSA accumulator updates:

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Our goal: design SNARKs that work directly over the integers.

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The blueprint for SNARKs over the integers

First, the difficulties

No inverses

Integers have no multiplicative inverses, which complicates extraction

Hard to commit

Commitment schemes for integers are scarce

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Proving statements over integers efficiently is difficult

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Modular PIOP (Mod-PIOP)
Polynomial interactive oracle proof: the information-theoretic part

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😞 Bad news: the Mod-PC cannot
The cryptographic part dictates the security and the assumptions of the scheme, and a normal PCS cannot simply be reused for it.

The Mod-PC and its landscape before this work

$\mathsf{Setup}(\lambda, N, B) \rightarrow (\mathsf{ck}, \mathsf{vk})$

$\mathsf{Commit}(\mathsf{ck}, f) \rightarrow \mathsf{cm}$

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The Mod-PC literature: surprisingly, only two constructions

Scheme	Built on	PQ	Unbounded integers	Extracts	Prover	Verifier	Proof
CHA24	Unknown-order groups	✗	✓	Integers	BN log N	log N · log² B	log² B
Zip, GWV+25	Hash-based codes	✓	✗	Rationals	N · B	√N · B	√N · B

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Qāren (this work) + TensorSwitch, BFRW25	Hash-based tensor codes	✓	✗	Rationals	BN log(BN)	log(B + N)	log(B + N)

The insight: evaluation as an inner product

We have a polynomial (multilinear or univariate), represented as a vector of coefficients or evaluations:

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$$f(\alpha) \cdot g^{\mathrm{rev}}(\alpha) = \underbrace{L}_{\sum_{i=0}^{N-2} c_i \, \alpha^i} + \underbrace{M}_{\langle \mathbf{f}, \mathbf{g} \rangle} \cdot \alpha^{N-1} + \underbrace{R}_{\sum_{i=N}^{2N-2} c_i \, \alpha^i}$$

Commitment and evaluation

Inspired by the last slide, we commit to a polynomial f in the following form:

Evaluate at α
$f(\alpha) \in \mathbb{Z}$

→

Binary decomposition
$\mathbf{a} = \mathrm{Bits}\big(f(\alpha)\big)$

→

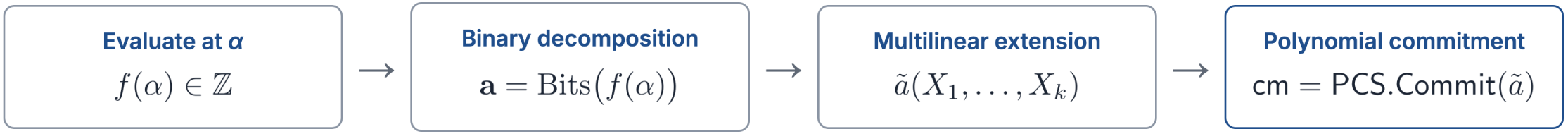
Multilinear extension
$\tilde{a}(X_1, \ldots, X_k)$

→

Polynomial commitment
$\mathsf{cm} = \mathsf{PCS.Commit}(\tilde{a})$

Commitment and evaluation

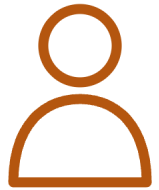
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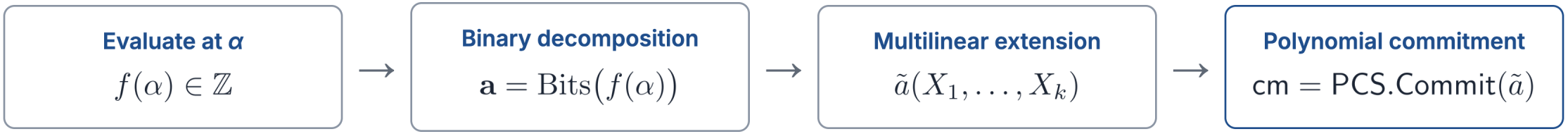
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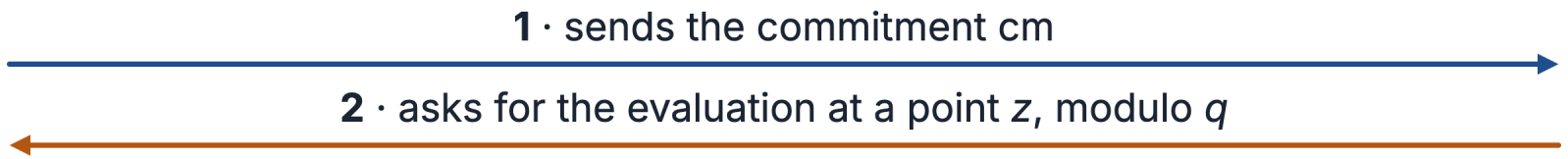


Verifier

Commitment and evaluation

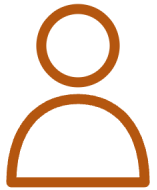
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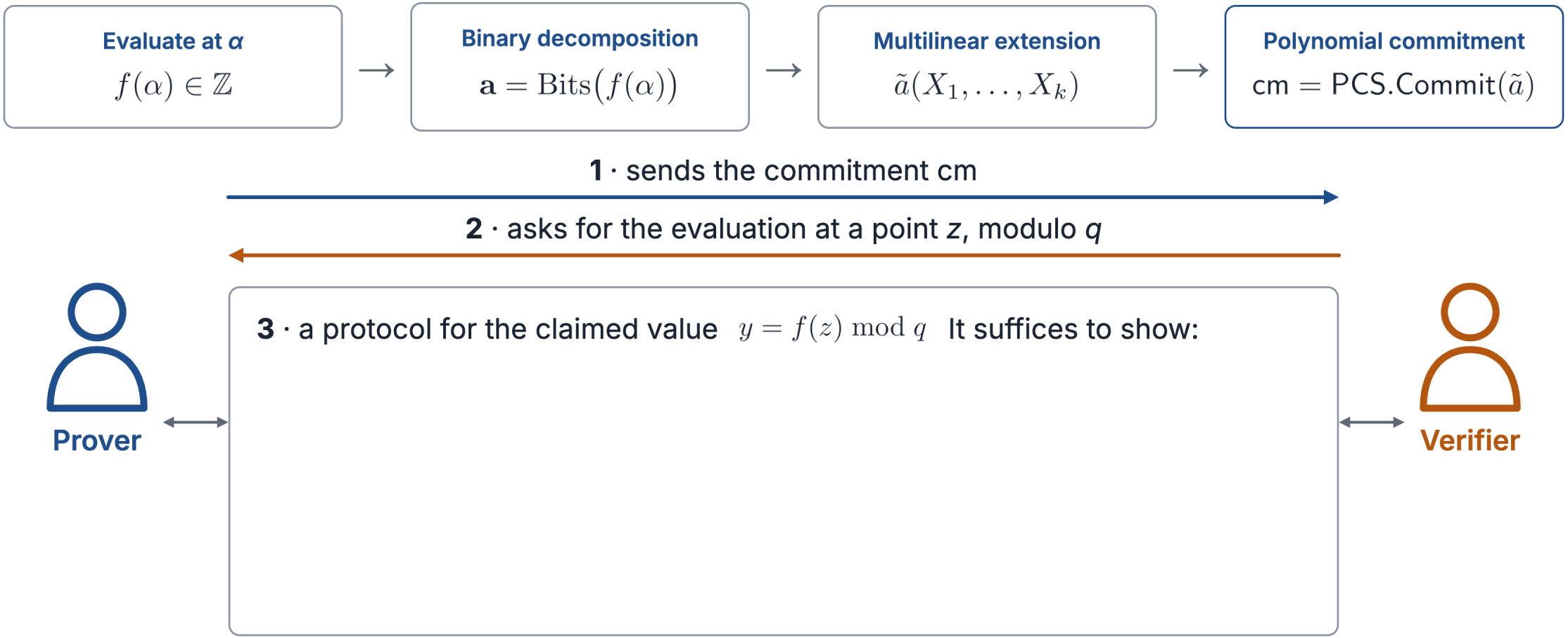
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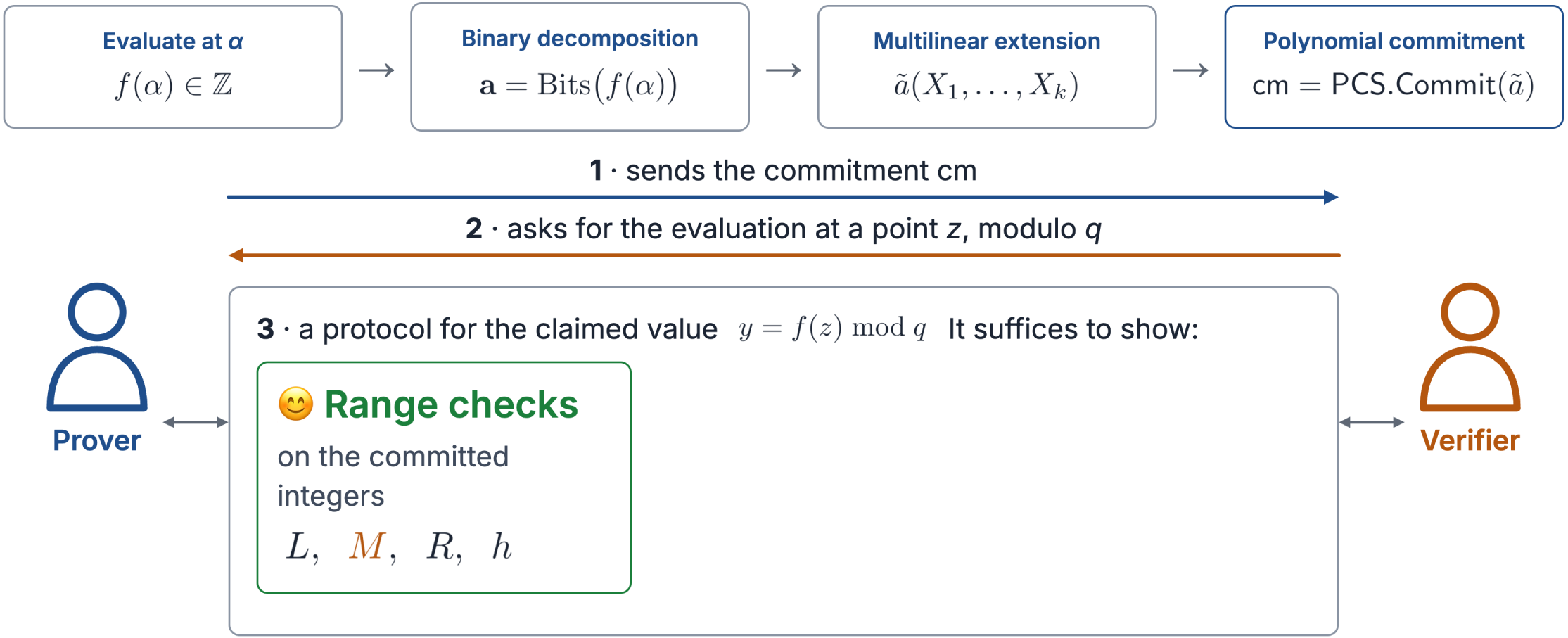
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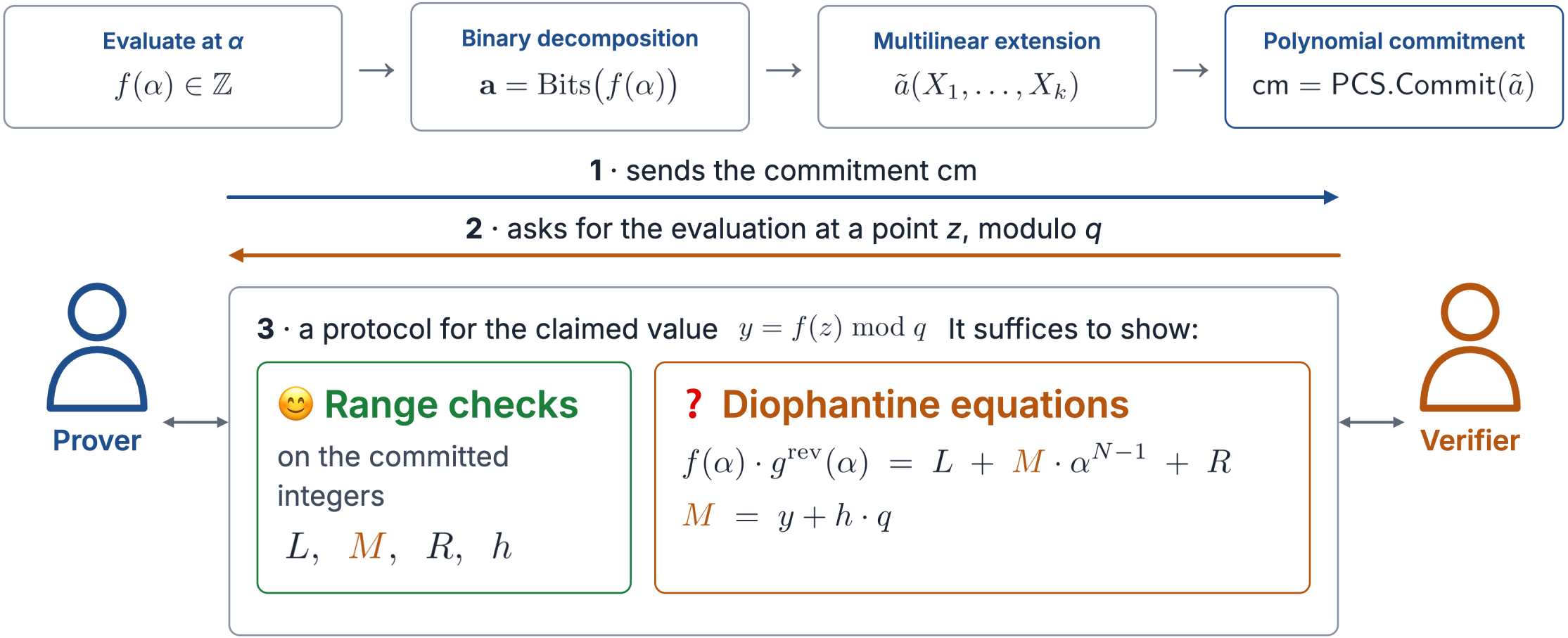
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Side contributions

MaSDAK: a protocol for *B*-bit Diophantine equations

Magnitude-**S**uccinct **D**iophantine **A**rgument of **K**nowledge

$$G(a_1, \ldots, a_t) = 0 \quad \text{over } \mathbb{Z}$$

Big-O omitted. *t*: number of integers. *s*: size of the equation. *B*: bit-length bound. GUO: groups of unknown order. RO: random oracle.

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PoKEMath, BMSS25	✓	✗	✓	✓	GUO	tB log B	λ(t + s log λ)	tλ
MaSDAK + TensorSwitch, BFRW25 (this work)	✓	✗	✓	✗	RO	tB log B	tλ + log tB + λ log λ	tλ + log tB

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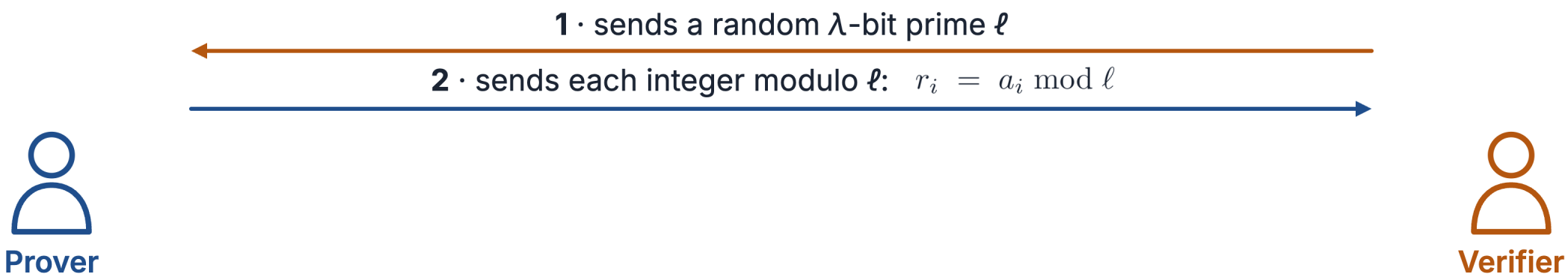
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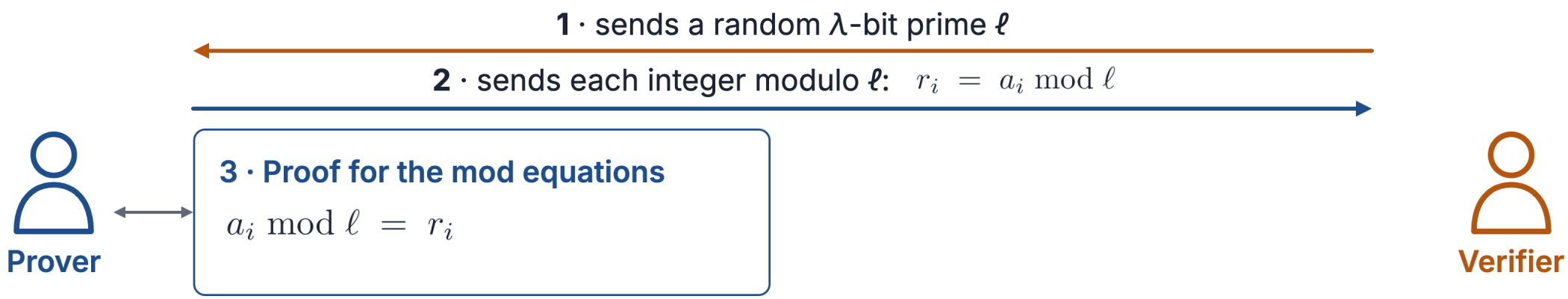
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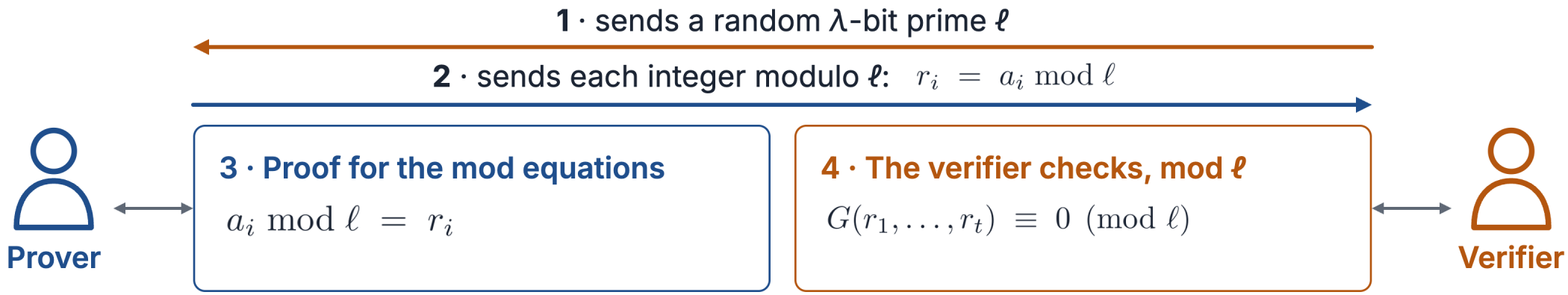
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MaSDAK + TensorSwitch, BFRW25 (this work)	✓	✗	✓	✗	RO	tB log B	tλ + log tB + λ log λ	tλ + log tB



Big-O omitted. *t*: number of integers. *s*: size of the equation. *B*: bit-length bound. GUO: groups of unknown order. RO: random oracle.

The modularity check

To check $a_i \bmod \ell = r_i$

Add: integer addition. Mul: integer multiplication. Big-O omitted. *t*: number of instances. *B*: bit-length bound.

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Scheme	PQ	ZK	Assumption	Add: prover	Add: verifier	Add: proof	Mul: prover	Mul: verifier	Mul: proof
LLNW18	✓	✓	SIVP	tB	tB	tB	tB^($\log_2 3$)	tB^($\log_2 3$)	tB^($\log_2 3$)
LNS20	✓	✓	R-SIS, R-LWE	tB	tB	tB	tB log B	tB	tB
KSSL21	✓	✓	M-SIS, M-LWE	tB	tB	tB	tB log B	tB log B	tB log B
This work + TensorSwitch, BFRW25	✓	✗	RO	tB	log tB	log tB	tB log B	log tB	log tB

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The hierarchy of our protocols

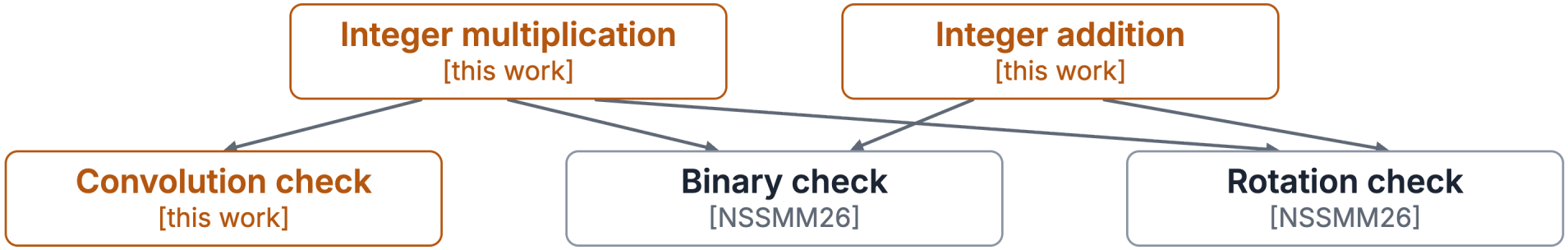
Integer multiplication and addition reduce, protocol by protocol, to sumcheck. Sumcheck leaves only evaluation claims.

Integer multiplication
[this work]

Integer addition
[this work]

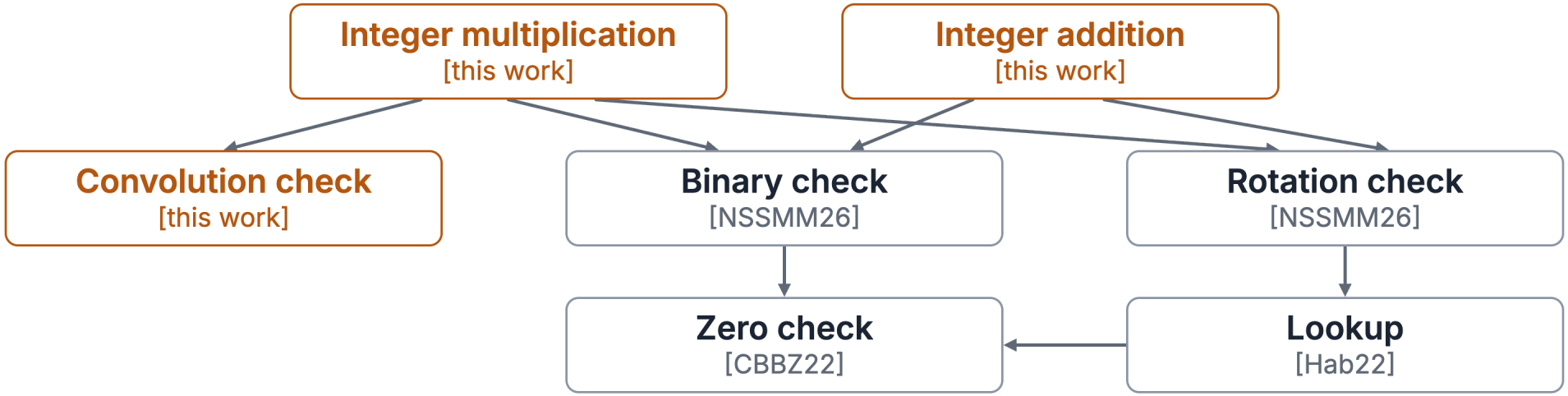
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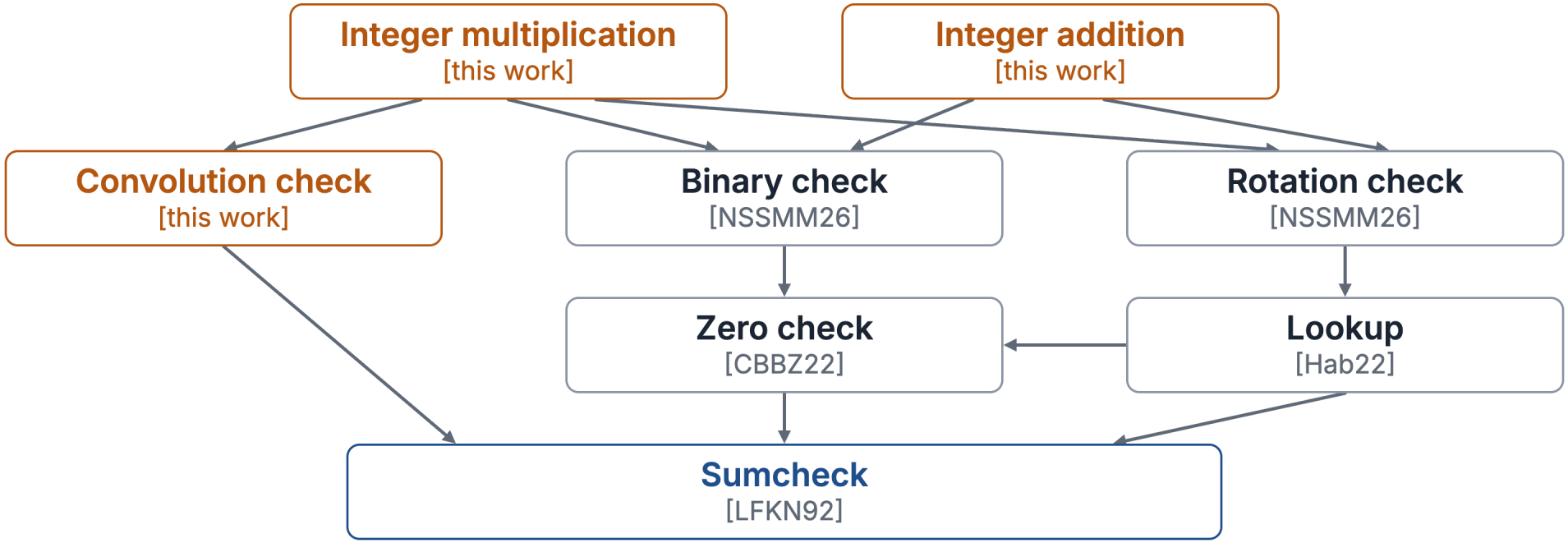
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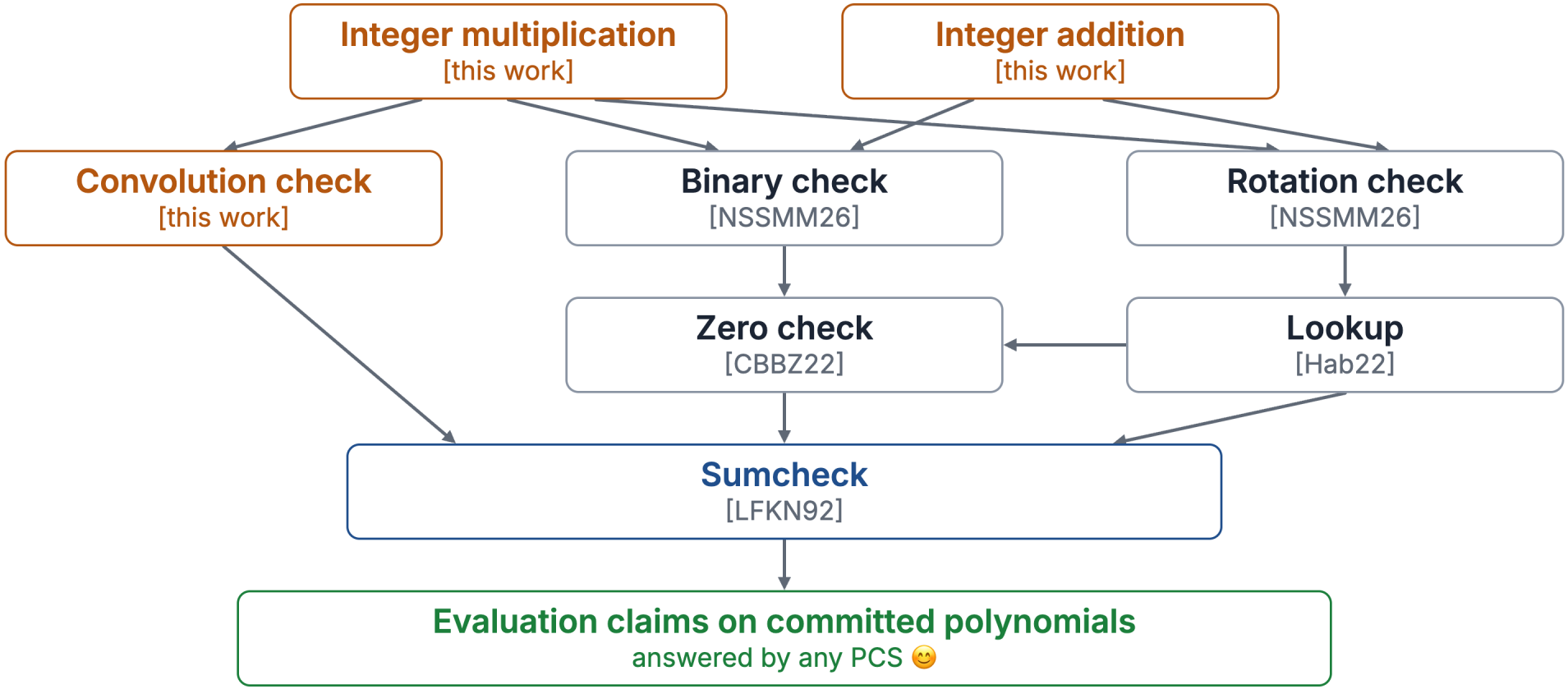
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From mod α to mod q: commitment switching

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We have $f(z) \equiv y + h \cdot q \pmod{\alpha}$ and we need $q \mid \alpha \implies f(z) \equiv y \pmod{q}$

But α is fixed at commitment time, before the verifier picks q.

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3 · commits again, and sends y: $\beta = q \cdot \alpha, \quad \mathsf{cm}' = \mathsf{Commit}(f(\beta))$



Prover



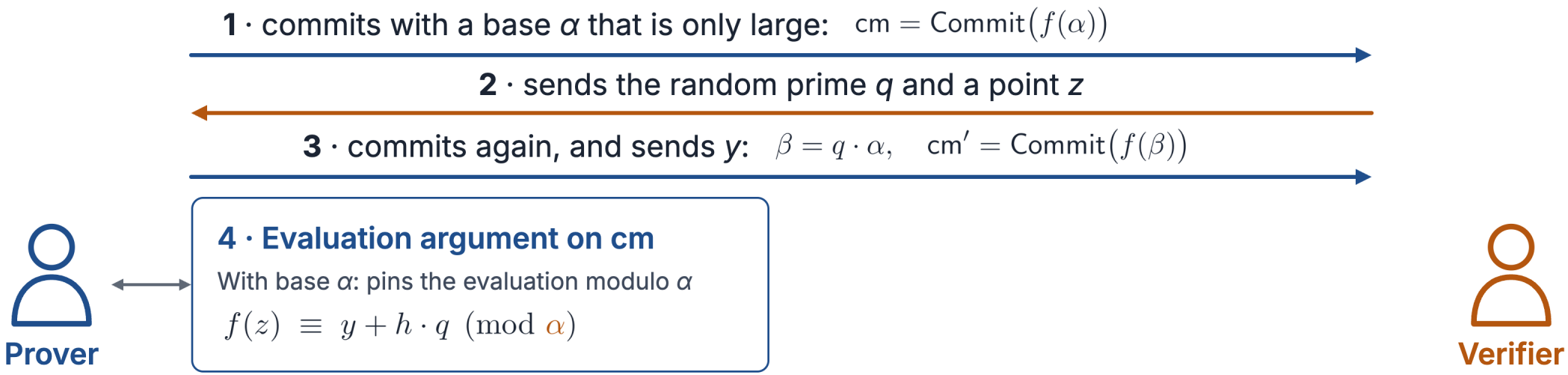
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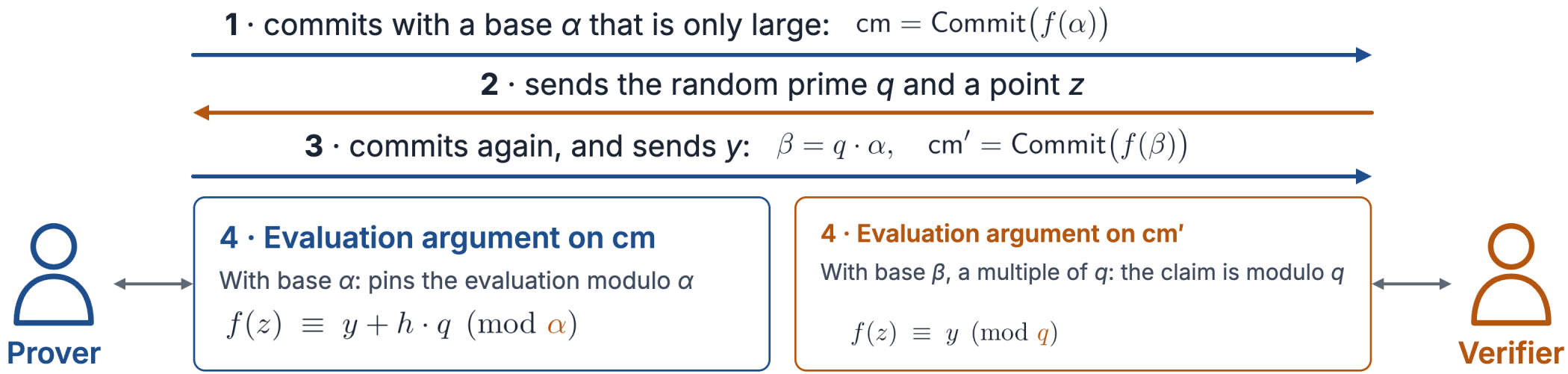


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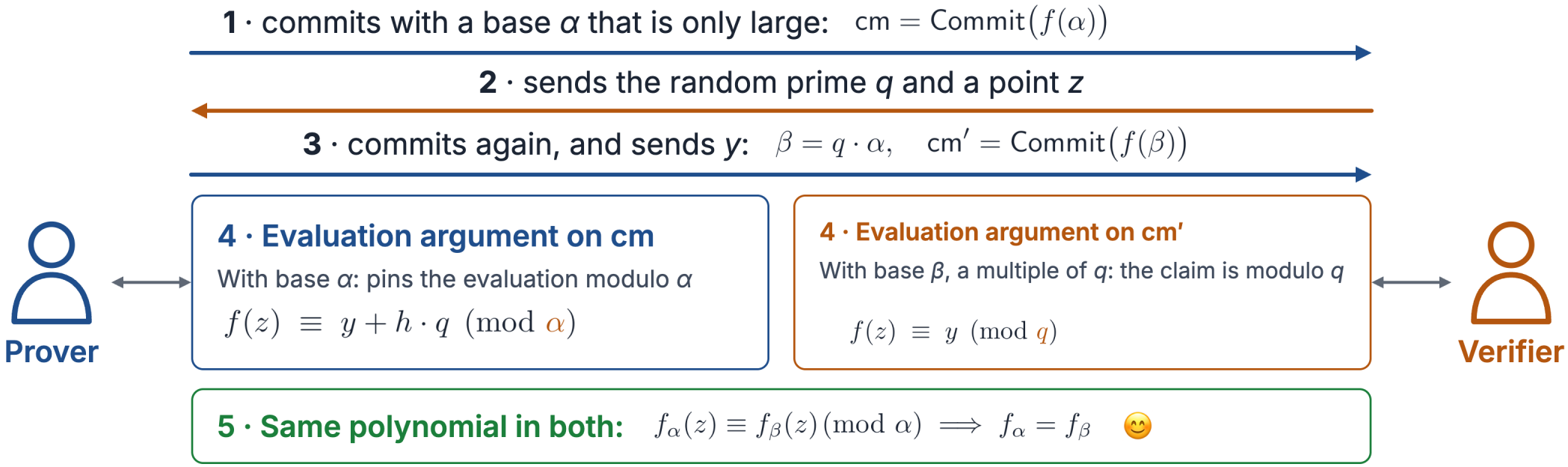


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Thank you!

Relaxed Modular PCS from Arbitrary PCS and Applications to SNARKs for Integers

Alireza Shirzad · Sriram Sridhar · Dimitrios Papadopoulos · Charalampos Papamanthou

eprint.iacr.org/2026/347

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These slides were created by AI under human supervision.